BedHopper

The Open Accommodation Protocol

Business Model Document

Version 1.0

A community-owned, monetizable public infrastructure
for ultra-low-cost accommodation worldwide.

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Executive Summary

BedHopper is not a traditional company that owns listings or controls a centralized marketplace. It is an **open-source accommodation protocol** — a piece of public infrastructure, similar to email or the web itself. Anyone can run a BedHopper node (a city server, a hostel network, a university housing board). All nodes can optionally federate, sharing listings globally.

The official `BedHopper.com` instance is just one node — the largest, most trusted one — which funds the protocol's ongoing development and provides premium services to the ecosystem.

Core Principles:

- **No central platform dependency.** Hosts and guests are never locked into a single company's ecosystem.
- **Network effects without monopoly.** The more nodes join, the more valuable the entire ecosystem becomes, but no single entity owns it.
- **Trust is portable.** A guest verified on one node carries their reputation to any other node worldwide — similar to how Mastodon works for social media, but for accommodation.

The Problem with Existing Models

Airbnb: The Centralized Marketplace

- **Model:** Asset-light, two-sided online marketplace.
- **Revenue:** 15–20% commission from both hosts and guests.
- **Host dependency:** Hosts are entirely dependent on Airbnb for visibility, reviews, and bookings. If delisted, their business disappears.
- **Lock-in:** Reviews, reputation, and booking history are trapped inside Airbnb's walled garden.
- **No budget focus:** Cleaning fees, service charges, and algorithmic pricing have pushed even "budget" listings to $30–50/night.

OYO: The Franchise & Aggregator

- **Model:** Asset-light franchise and aggregator.
- **Revenue:** Franchise fees, revenue share from properties, and booking commissions.
- **Host dependency:** Hotels must rebrand, follow OYO's operational standards, and surrender significant control over pricing and customer relationships.
- **Lock-in:** OYO controls branding, operations, and the customer relationship. Properties cannot easily leave.
- **Not for micro-hosts:** Individuals with a spare couch or single room are completely excluded.

The Gap: No Public Infrastructure for Budget Accommodation

Neither model serves the **sub-$15/night market**. Neither is open, portable, or community-owned. The budget accommodation space — couches, shared rooms, hostel dorm beds, service-exchange stays — is fragmented across Couchsurfing (no payments, safety issues), Workaway (paid membership, no booking system), and small hostels (invisible on major OTAs).

BedHopper fills this gap with an entirely new model.

The BedHopper Business Model: The Open Accommodation Protocol

Three-Sided Architecture

Unlike Airbnb (two-sided: guest–host) and OYO (two-sided: franchise–customer), BedHopper is a **three-sided protocol** with travelers, hosts, and **node operators**.

lightbg **Side**	**Who**	**Role & Benefit**
Travelers	Budget travelers, students, digital nomads, migrant workers	Find and book the cheapest beds globally. One portable identity across any BedHopper node. Service-share options for free stays.
Hosts	Individuals with a spare couch, hostels with empty dorm beds, guesthouses, capsule hotels	List beds at any price or offer service-exchange. No commission to the protocol. Own their data and reviews. Can switch nodes or run their own.
Node Operators	City cooperatives, hostel chains, universities, NGOs, housing collectives	Run a BedHopper node for their community. Custom rules (price caps, verification standards). Keep 100% of local revenue. Optionally federate to share listings globally.

Table 1: Three-Sided Protocol Architecture

How It Works: Federation, Not Aggregation

1. A node operator installs BedHopper (free, open-source) on their server.
2. They configure local rules: price caps, verification requirements, supported languages, payment methods.
3. Hosts in that city or network list their beds on the local node.
4. Travelers search and book beds within that node.
5. If the node operator *chooses* to federate, their listings become visible across the entire BedHopper network, and they receive bookings from travelers on other nodes.
6. The official BedHopper node provides the federation infrastructure and premium services — but is never mandatory.

This is the same architecture as **Mastodon** (social media), **Matrix** (messaging), and **WordPress** (publishing) — open protocols where anyone can participate, but convenience services generate sustainable revenue for the core development team.

Revenue Model: Monetizing Infrastructure, Not Hosts

Unlike Airbnb and OYO, BedHopper earns **zero commission from hosts**. Revenue comes entirely from value-added infrastructure services for node operators and premium features for travelers.

Stream 1: Node Operator Licensing (B2B — Infrastructure as a Service)

Node operators who don't want to self-host can pay for a managed, white-label BedHopper instance with guaranteed uptime, automatic updates, and optional global federation.

lightbg **Tier**	**Price**	**What's Included**	**Target Customer**
Community	Free (open-source)	Self-hosted, community forum support, manual federation setup	Tech-savvy collectives, small hostels
Standard	$99/month	Managed hosting, automatic updates, basic analytics, global federation activated	City cooperatives, small hostel chains
Enterprise	$499–$2,000/month	Custom domain, white-label branding, advanced analytics, priority support, dedicated federation relay, API access, bulk onboarding	Universities, large hostel networks, housing NGOs, government refugee programs

Table 2: Node Operator Tiers

This is the WordPress model: software is free; managed hosting and premium features generate recurring revenue.

Stream 2: Traveler Trust Passport (B2C)

A traveler who verifies their identity and builds a reputation on any BedHopper node can carry that **Trust Passport** to any federated node worldwide. This is a premium subscription.

lightbg **Feature**	**Free Tier**	**Trust Passport ($3/month or $25/year)**
Book beds on any node	✓	✓
ID-verified badge	×	✓
Portable reputation (cross-node)	×	✓
Service-share deposit waived	×	✓
Priority booking on high-demand listings	×	✓
24/7 premium support	×	✓

Table 3: Traveler Trust Passport Tiers

This creates a recurring revenue stream that scales with active travelers, independent of booking volume.

Stream 3: Federation Relay & Data Services (B2B)

Nodes that choose to federate (share listings globally) can use the official federation relay for reliable, high-performance sync. Additionally, anonymized market intelligence reports are sold to institutional customers.

lightbg **Service**	**Description**	**Price**
Federation relay (Standard)	Sync up to 10,000 listings globally with guaranteed uptime	$49/month
Federation relay (Unlimited)	Unlimited listings, priority sync speed, dedicated support	$149/month
Market Intelligence Reports	Anonymized demand trends, pricing benchmarks, city-level occupancy data	$99/report or $199/month
API Access	Third-party integration for insurance providers, payment gateways, travel agencies	Pay-per-call or monthly tier

Table 4: Federation Relay & Data Services

Comparison: BedHopper vs Airbnb vs OYO

lightbg **Aspect**	**Airbnb**	**OYO**	**BedHopper**
Asset ownership	None (marketplace)	None (franchise/aggregator)	None (protocol)
Host relationship	Dependent users on a centralized platform	Franchisees or aggregated inventory, OYO-controlled	Co-owners of a public network; can leave and run their own node anytime
Revenue extraction	15–20% commission from host + guest	Revenue share from franchise, booking commission	**Zero commission from hosts.** Revenue from infrastructure services and traveler subscriptions
Platform lock-in	High — reviews and data trapped in Airbnb	High — OYO controls branding and operations	**Zero lock-in** — portable identity and reviews; any node can fork and leave
Governance	Corporate, closed	Corporate, closed	Open protocol; node operators govern collectively (Linux Foundation model)
Trust model	Centralized verification, walled garden	Centralized quality control	Decentralized but interoperable Trust Passport
Budget focus	No — prices inflated by fees	No — standardized, branded rooms	**Yes** — built for sub-\$15/night accommodation

Table 5: Competitive Comparison

Why This Model Wins

For Hosts: Zero Commission, Full Control

Hosts never pay a percentage of their earnings. They own their listing data, reviews, and guest relationships. If a node operator raises fees or changes rules they dislike, hosts can switch to another node or run their own — their reputation follows them. **This is impossible on Airbnb or OYO.**

For Travelers: One Identity, Anywhere

A traveler verifies once and builds a single reputation. That Trust Passport works on any BedHopper node globally. They are never locked into one platform. Service-share options make travel accessible even with zero cash.

For Node Operators: Run a Business on Public Infrastructure

Universities, hostel chains, and housing NGOs get a production-ready accommodation platform without building from scratch. They keep 100% of their local revenue. The official node handles the hard parts — global sync, security updates, identity verification — as a paid service.

For the Protocol: Unstoppable Network Effects

The more independent nodes launch, the more valuable the network becomes. But no single node controls it. Even if the official BedHopper company disappears, the software and data survive across hundreds of independent nodes. **This is infrastructure, not a startup.**

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Financial Projections (3-Year Outlook)

Revenue Projections

lightbg **Stream**	**Year 1**	**Year 2**	**Year 3**
Node Operator Subscriptions	$36,000	$180,000	$720,000
Trust Passport Subscriptions	$18,000	$120,000	$450,000
Federation Relay Fees	$12,000	$72,000	$240,000
Market Intelligence	$6,000	$36,000	$120,000
API Access	$3,000	$24,000	$90,000
Total Revenue	**$75,000**	**$432,000**	**$1,620,000**

Table 6: Revenue Projections (Conservative Estimates)

Key Assumptions

- Year 1: 30 paying node operators, 500 Trust Passport subscribers, 10 federation relay clients.
- Year 2: 150 paying node operators, 3,000 Trust Passport subscribers, 60 federation relay clients.
- Year 3: 600 paying node operators, 15,000 Trust Passport subscribers, 200 federation relay clients.
- Gross margins remain high (70–80%) as infrastructure costs scale sub-linearly.

The Core Philosophy

Build Once, Monetize the Convenience, Own Nothing That Hosts Own

BedHopper does not compete with Airbnb on being a better marketplace. It competes on being a fundamentally different category: **public infrastructure for accommodation.**

> **The Model in One Sentence**
>
> Build the open-source infrastructure for ultra-low-cost accommodation, let anyone run it, and monetize the convenience layer — managed hosting, portable trust, and federation — without ever charging hosts a commission.

This is not a marketplace business. This is not a franchise business.
This is a **public utility business model** — like the postal service, but for cheap beds.

Build once. Deploy everywhere. Monetize the value layer.
The core remains free and open forever.

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